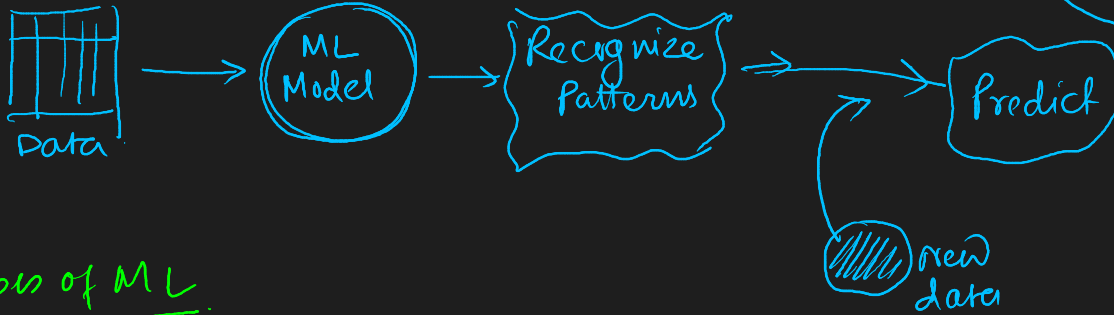
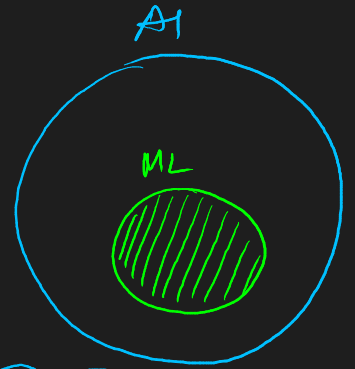


Module-1

ML

Q. What is ML? types?

ML is a branch of AI that enables computers and improve their performance without being explicitly programmed.



Types of ML

- ① **Supervised**:
- Train under supervision.
 - Model is trained using labelled data.
 - The input and correct output is given while training.
 - Used for prediction & classification task.
 - Email spam detection, house price prediction.
 - Algorithms: Linear Regression, Logistic Regression, SVM, etc.

- ② **Unsupervised**:
- No supervision.
 - Model works with unstable data and finds hidden patterns or relationships.
 - Used for grouping or data analysis.
 - Customer Segmentation, Recommendation System.
 - Algos - K-Means, PCA.

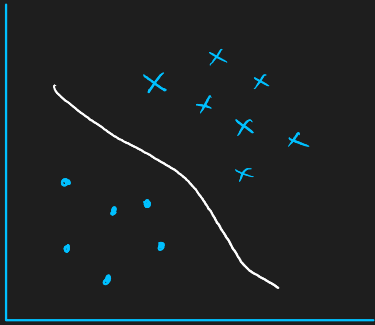
- ③ **Reinforcement** -
- An agent learns by interacting with environment.
 - After taking correct step the environment gives reward to agent.
 - If wrong steps taken - penalties are there.
 - Robotics, Gaming, Self-Driving Cars, Robot Navigation.
 - Agent improves its action with trial and error.

- ④ **Semi-Supervised**:
- Uses both labelled/unlabelled while training.
 - helpful when labelled data is limited.
 - Image Recognition and Medical diagnosis.

Q.2. Supervised vs Unsupervised.

Supervised

- Labelled data.
- correct o/p is known already.
- Learns by comparing predictions with actual answers.
- used for classification/prediction
- Eg: Predict House Price
- Algo: Linear Regression, DT



Unsupervised

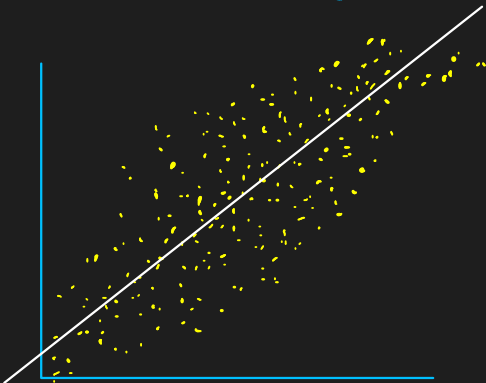
- Unlabelled data.
- Correct o/p not known.
- Learns by finding hidden patterns in data.
- Clustering/Grouping.
- Eg: Grouping customers by behaviour.
- Algo: K-Means, PCA.



Q.3. Regression vs. Classification

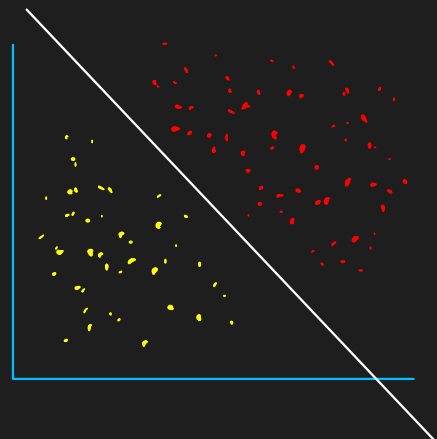
Regression

- Predicts continuous values.
- o/p - Numeric
- used when the ans can have many values.
- Eg: Predicting House Price.
- Eg o/p: ₹40,000
- Algo: Linear Regression



Classification

- Classes/categories.
- Discrete (Labels) → Y/N.
- used when answer belongs to a fixed gr.
- Eg: Predicting Spam/not.
- Eg. o/p: Yes/No
- Algo: Logistic Regression.



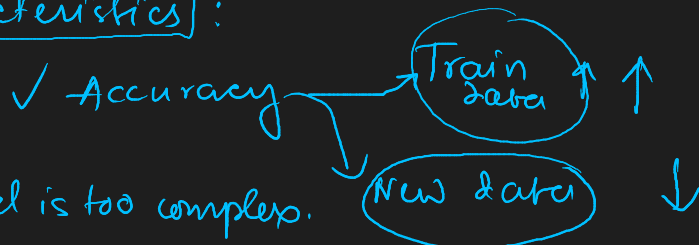
2.4. Overfitting & Underfitting.

Overfitting:

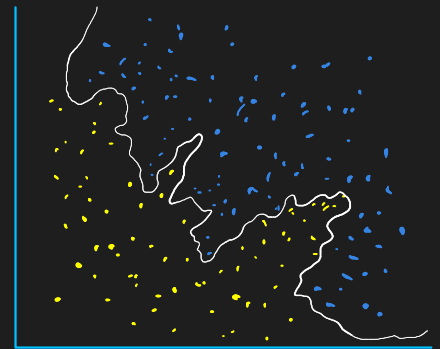
- Happens when the ML Model learns the training data too well including noise and unnecessary details.
- Performs well on training data.
- Performs poorly on new/unknown data.

• **Eg**: A student memorizes all answers instead understanding the concept. In exam the questions changed slightly, the student cannot answer properly.

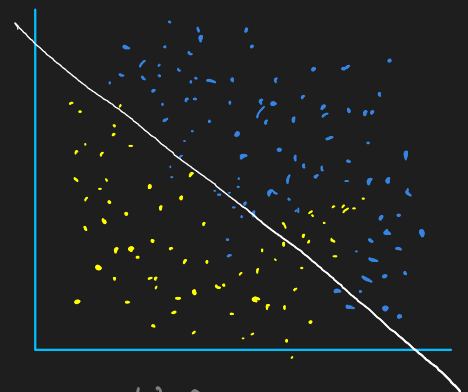
Characteristics:



- **Cause**
- Train data is too small
 - Epochs ↑
 - Noisy data - complex Model



Overfitting



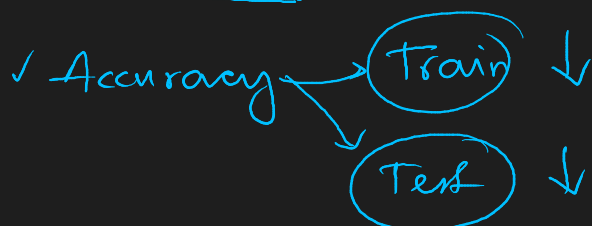
Underfitting

Underfitting:

- When the model is too simple & cannot learn the patterns from the data properly.
- Performs poorly on both training & testing data.

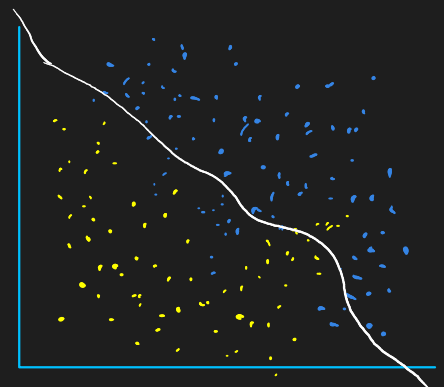
• **Eg**: A student studies very little & does not understand the concepts properly.

Characteristics:



✓ Model is too simple.

- **Cause**
- Insufficient training.
 - too few features. - High bias.



Perfect.

Q.5. What is cost fn?

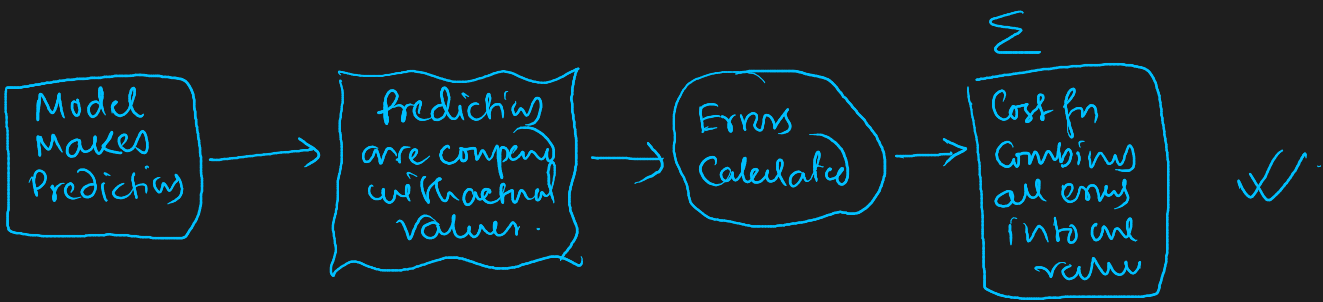
Cost function is a mathematical fn used in ML to measure how wrong the model's predictions are. Actually it tells us how much error is +nt in the model.

The main goal of training a Model is Minimize cost fn value.

{ cost ↓ → Model Performing Well.
cost ↑ → Model is performing Poorly. }

cost fn measures the overall error of the model on the entire training dataset.

Cost fn = avg(all prediction errors)



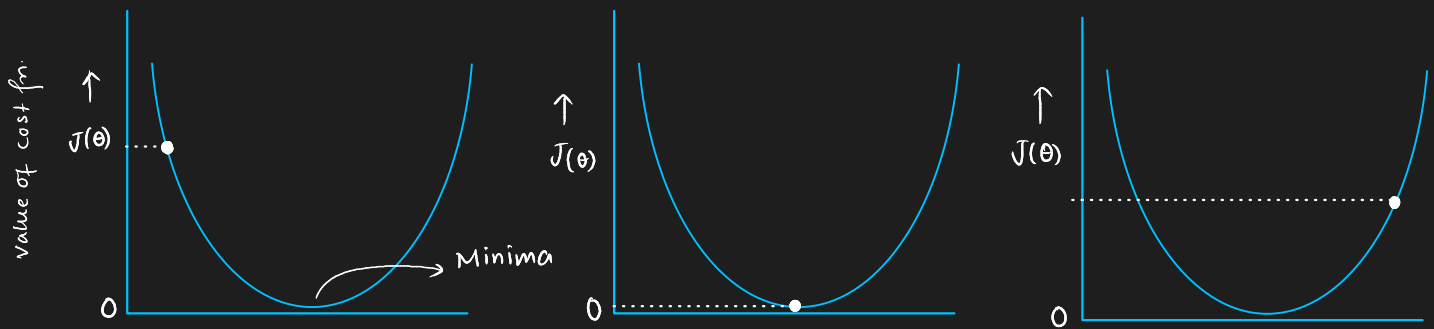
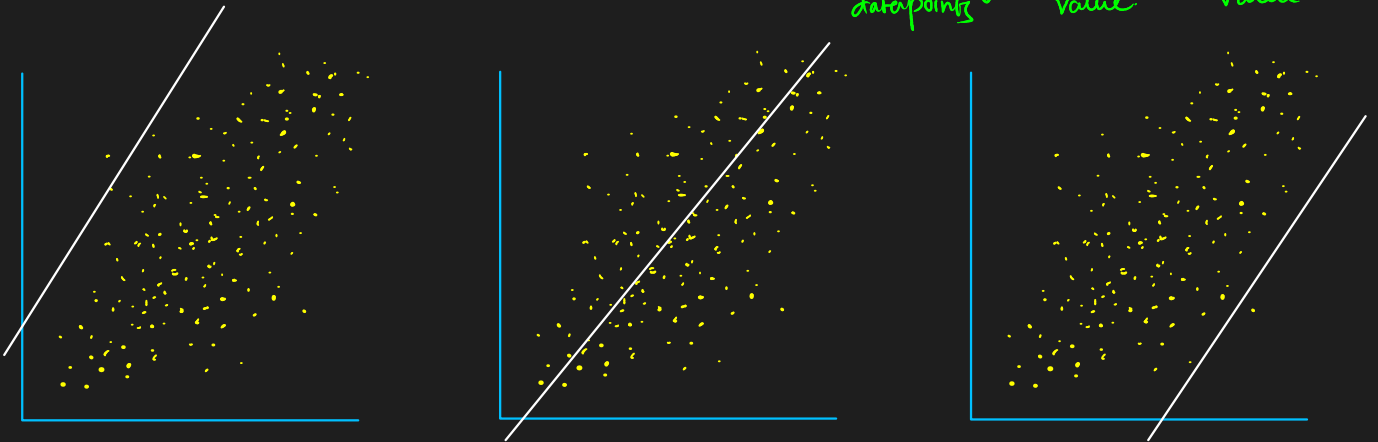
[Eg]: for regression:

A common cost fn is MSE:

$$J(\theta) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

of training datapoints
actual value
Predicted value

Scatterplot & Best Fit Line.



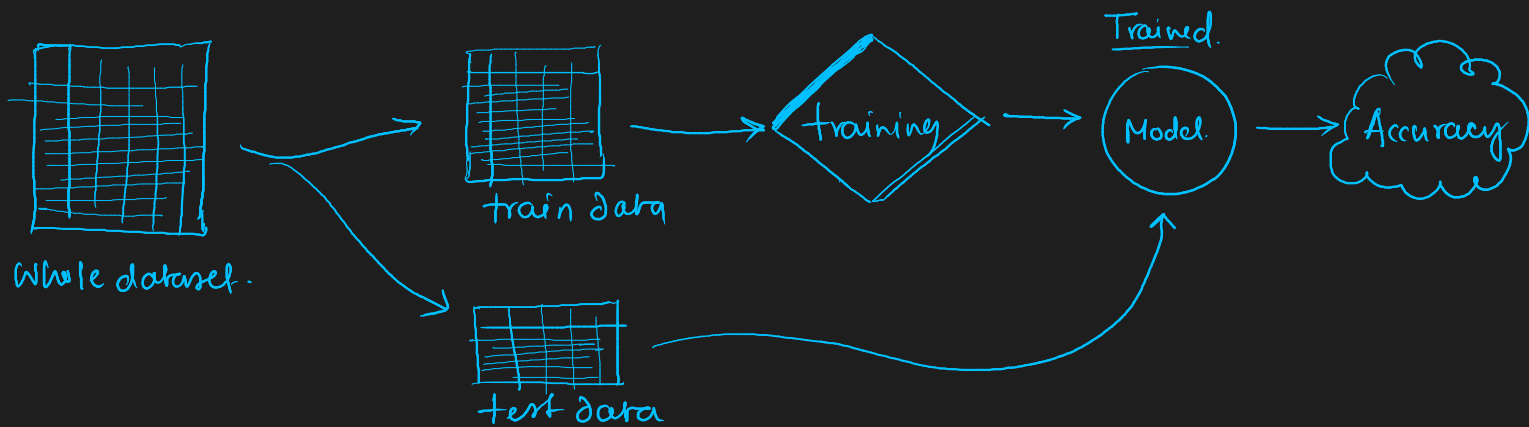
Q.6: Training & Test set.

Training Set : • A portion of data used to train the ML model.
• Model learns Rules, Patterns from this data.

Purpose → • Teach the model.
• Adjust weights & biases.
• Improve predictions.

Test Set : • Used to check the performance of the trained Model.
• The model has never seen the data before.

Purpose : → • Evaluate accuracy.
• Check generalization ability.
• Detect Overfitting/Underfitting.



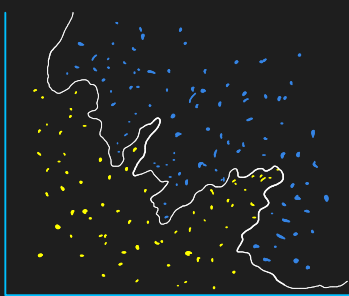
Q.7: Overfitting - Underfitting tradeoff in Regression -

- Explain Overfitting.
- Explain Underfitting.

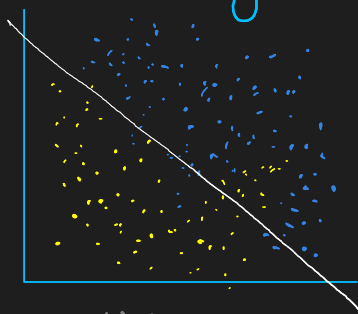
- **Tradeoffs** : • Model complexity $\uparrow \rightarrow$ Underfitting $\downarrow$
Overfitting $\uparrow$

The goal is to find a balance where -

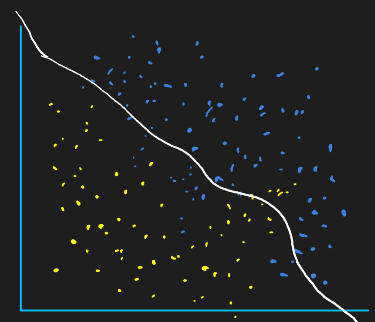
- Training Error $\downarrow$ Testing Error $\downarrow$
- Model generalizes well for new data.



Overfitting



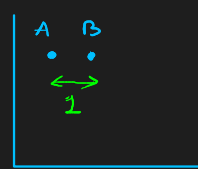
Underfitting



Balanced.

Q.8. Euclidean dist. b/w $A(1,3)$, $B(2,3)$?

$$d_e = \sqrt{(1-2)^2 + (3-3)^2} = 1$$



Q.9. Same as Q.6.

Q.10. bias-variance tradeoff.

↳ balance b/w underfitting & overfitting in ML. → same as Q.7

Supervised Learning

Q. What is LR algorithm? How to interpret a LR Model.?

↓
Linear Regression.

Q. Numerical & or simple LR?

<https://anikchandml.hashnode.dev/8-linear-regression>

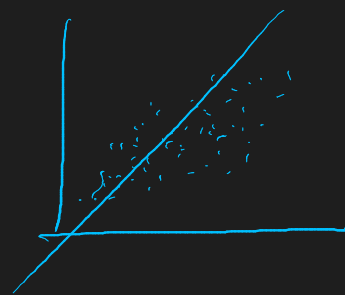
Linear Regression: A supervised ML algorithm used to predict a continuous numerical value by finding a linear relationship between i/p and o/p variables.

- Fits a straight line to the data.

Equation:

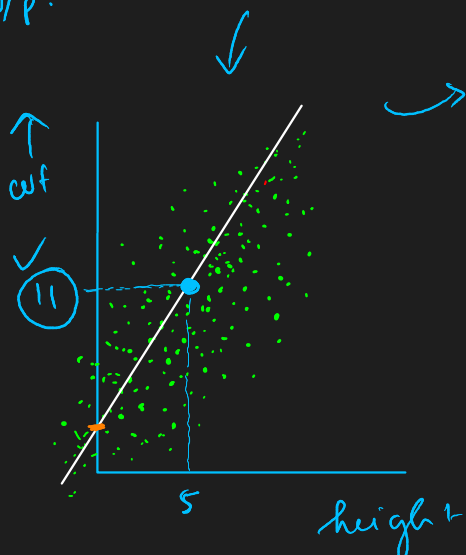
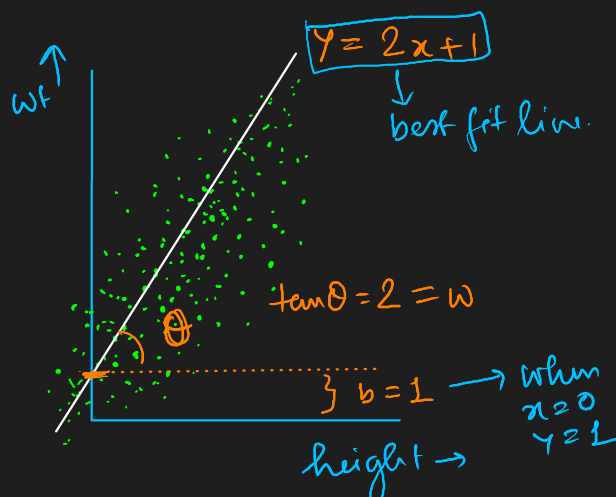
$$y = wx + b$$

Annotations:
- y : Predicted o/p
- w : weight
- x : input
- b : bias



Interpretation: Suppose we have trained a LR model on a dataset having height i/p & weight o/p.

test a new data. (height = 5)



$$y = 2 \times 5 + 1 = 11 \checkmark$$

So, if the height is 5 unit the wt. will be 11 unit.

$w = \text{slope} = 2 \rightarrow$ for every 1 unit change in x , y changes by 2

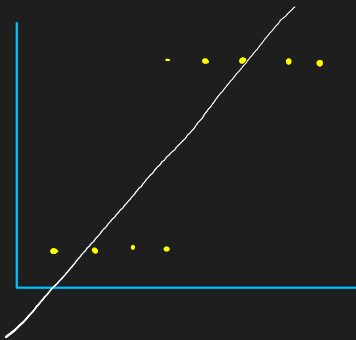
Q. What is Logistic Regression? What is Logistic or sigmoid fn? Write its role in Logistic Regression.
Why does LR comes under the category of classification?

<https://anikchandml.hashnode.dev/9-logistic-regression>

Logistic vs. Linear Regression.

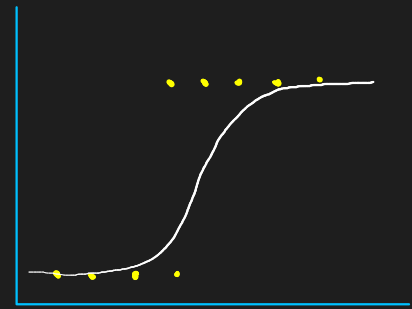
Linear Regression

- used for regression
- Predicts a continuous numerical value.
- $o/p \rightarrow$ any Real No
- fits a straight line



Logistic Regression

- used for classification.
- Predicts a class/category.
- between 0 & 1 $[0, 1]$
- fits a S-shaped curve.



KNN:

1. Which machine learning algorithm is called lazy learner and why?
2. Write the advantages and disadvantages of KNN algorithm.
3. Why is KNN a non-parametric Algorithm?
4. Why is the odd value of "K" preferred over even values in the KNN algorithm?
5. How does the KNN algorithm make the predictions on the unseen dataset?
6. Write down the steps of KNN algorithm.
7. Numerical question on KNN.

<https://anikchandml.hashnode.dev/6-knn>

Naïve Bayes Classifier:

1. Define Bayes theorem. What is the basic assumption in Naïve Bayes classifier?
2. Discuss the advantages and limitations of Naive Bayes algorithm.
3. Numerical question

<https://anikchandml.hashnode.dev/7-naive-bayes-algorithm>

Decision Tree:

1. Explain the terms Information Gain and Gini index with the help of an example.
2. Construct a decision tree Training data set is given.
3. What is a decision tree? Explain with an example.
4. Write down the ID3 algorithm.
5. Advantages and disadvantages of Decision Tree.
6. Numerical question

<https://anikchandml.hashnode.dev/3-decision-trees>

<https://anikchandml.hashnode.dev/4-id3-algorithm-in-decision-tree>

<https://anikchandml.hashnode.dev/5-c4-5-algorithm-in-decision-tree>

SVM:

1. Explain the working principle of support vector machine algorithm.
2. Write the difference between linear and non-linear SVM.
3. What is the Euclidian distance between two data points?
4. Numerical question (Linear SVM)

<https://anikchandml.hashnode.dev/10-svm>

1. Confusion matrix will be given. Find out the accuracy, misclassification rate, sensitivity and other metrics.

	Predicted +ve	Predicted -ve
Actual +ve	TP	FN
Actual -ve	FP	TN

$$\left. \begin{aligned} TP &= \text{True +ve} \\ FN &= \text{false -ve} \\ FP &= \text{+ve} \\ TN &= \text{True -ve} \end{aligned} \right\}$$

$\begin{array}{c} \text{T/F} \\ \uparrow \\ \text{Predict}^n \end{array}$ $\begin{array}{c} \text{class} \\ +/- \\ \uparrow \\ \text{Predicted} \end{array}$

$$\text{Accuracy} = \frac{TP + TN}{TP + FN + FP + TN}$$

$$\text{Misclassification Rate} = \frac{FP + FN}{TP + FN + FP + TN}$$

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

(Recall)
(True +ve Rate)

$$\text{Specificity} = \frac{TN}{TN + FP}$$

(True -ve Rate)

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$F1 \text{ score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

2.

		Predicted Class			
		System A		System B	
Actual Class		Negative	Positive	Negative	Positive
	Negative	80	20	85	15
	Positive	25	75	30	70

Find out the number of False Positives (FP) and False Negatives (FN) for both systems.

→ Ans:

$$\left. \begin{aligned} FP(A) &= 20 \\ FN(A) &= 25 \end{aligned} \right\}$$

$$\left. \begin{aligned} FP(B) &= 15 \\ FN(B) &= 30 \end{aligned} \right\}$$

Imp:

TP = True Positive

Means.

Model prediction is True.

Model Predicted +ve.

Model is predicted the output +ve and actually that is +ve. Model's prediction is true.

FN = False Negative

Means.

Model prediction is False.

Model predicted value -ve.

Model is predicted the opp -ve and that is actually wrong. Actually the opp should be +ve. Model gave a False predictⁿ.

FP = False Positive

Means.

Model prediction is False.

Model predicted value $\rightarrow$ +ve.

Model is predicted the opp +ve. and that is wrong. Actually opp should be -ve. Model gave False predictⁿ.

TN = True Negative

Means

Model prediction is True

Model predicted value = -ve

Model gave the output -ve. and that is correct (True). Actually that is -ve. Model gave True predictⁿ.

Module - 4

Q.1. Define time series and time series data?

Time Series: Sequence of datapoints collected or recorded over time intervals. The observations are arranged in chronological order.

- Eg.:
- Daily Temperature
 - Stock price
 - Monthly Sales
 - Hourly Electricity usage.

A time series can be represented as: $X = \{x_1, x_2, x_3, \dots, x_n\}$

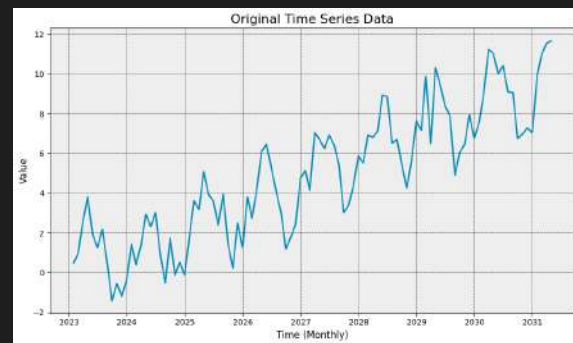
where, - x_1, x_2, x_3 are observation.

- each value is recorded at a specific time.

TS data: the data that is collected over the time continuously.

- Unlike normal dataset here the order of the observations are very very important.

Eg. of Time series data →



Q.2. Different Methods for TS forecasting?

① **Naive Forecasting**: • Simplest forecasting Method.

- Future value is assumed to be equal to prev. value.

$$\hat{y}_{t+1} = y_t$$

② **Moving Avg (MA)**: • Forecasting is calculated using average of prev. observations.

$$\hat{y}_{t+1} = \frac{y_t + y_{t-1} + \dots + y_{t-n+1}}{n}$$

- smooths noise, identifies trends.

③ **Weighted Moving Avg (WMA)**: • Recent observations are given higher importance.

$$\hat{y} = w_1 y_t + w_2 y_{t-1} + \dots + w_n y_{t-n+1}$$

where, $(w_1 + w_2 + \dots + w_n = 1)$

11) ML Methods :

- Linear Regression
- Random Forest
- XGBoost
- LGBM
- SVR

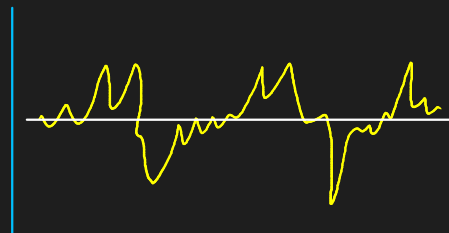
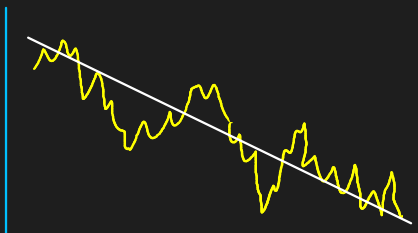
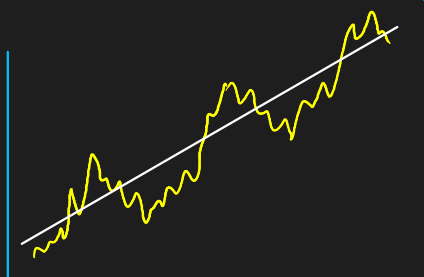
12) DL Methods :

- RNN
- LSTM
- GRU
- Transformers

8.3 Components of TS.

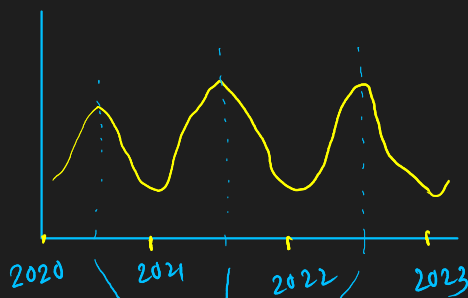
① **Trend (T)** : Represents the long term movement of data over the time.

- It shows whether the data is generally increasing, decreasing or constant.

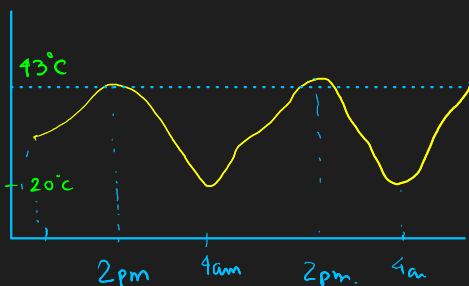


② **Seasonal Variation (S)** : Seasonality refers to patterns that repeat at fixed time intervals.

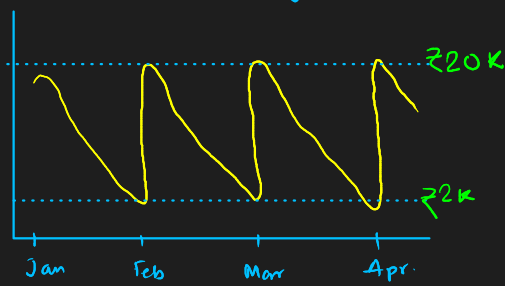
- Patterns occur like : Daily, Monthly, Quarterly, Yearly.



Mid of the yr. (Jun-Jul)
Peak icecream sale.
(Yearly)



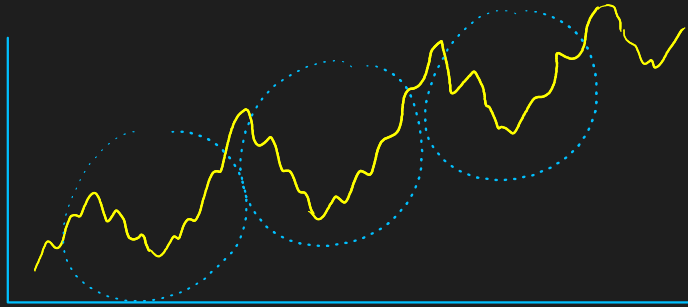
(daily whole day temp.)



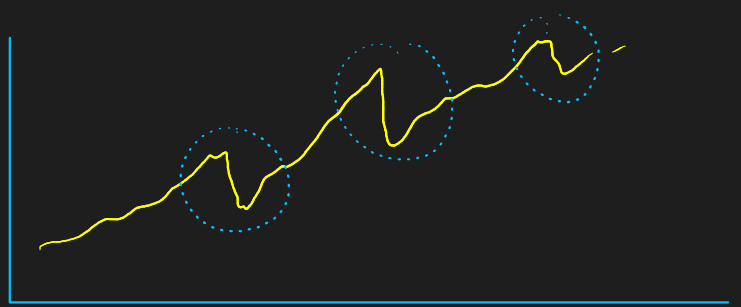
✓ Monthly Bank Balances
in the ending Bank Bal. ↓
Starting of any Month
Bal Max.

③ **Cyclical Variation (C)** : This represents long term fluctuations occurring over several years.

- Unlike seasonality cycles do not occur at fixed intervals.

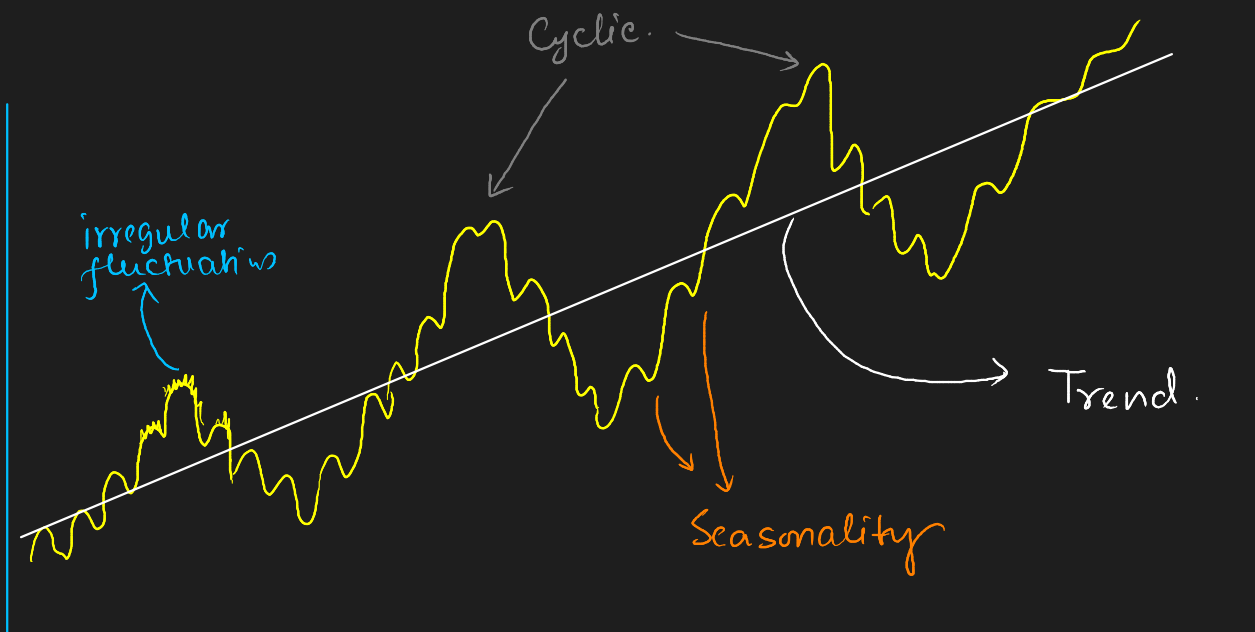


Stock Market cycles.



gold price over the yrs.

- ④ Irregular Variation/ Noise (I):
- Irregular variation consists of random and unpredictable changes in data.
 - Occurs due to unexpected events:
 - Flood
 - Pandemic
 - Earthquake.
 - Sudden stock Market crash.



Combined TS Model $\rightarrow$

$$Y = T + S + C + I$$

Q.4. SMA with example.

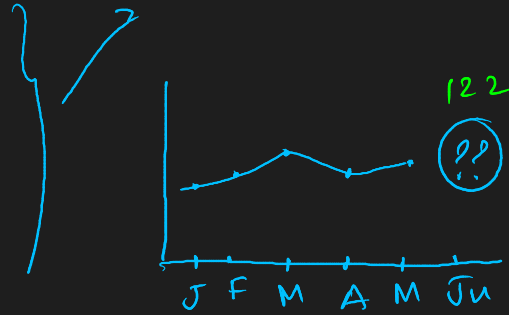
- SMA (Simple Moving Avg.) is a forecasting model used in Time series Analysis.
- Predicts future values by taking avg. of previous observations.
- Helps to $\rightarrow$
 - smooth fluctuations.
 - Reduce Noise
 - identify trends.

$$Y_{t+1} = \frac{Y_t + Y_{t-1} + \dots + Y_{t-n+1}}{n} \quad \checkmark$$

Eg.

Month	Sales
Jan	100
Feb	120
Mar	140
Apr	120
May	130

5 Month MA Forecast for June.



$$\frac{100 + 120 + 140 + 120 + 130}{5} = 122 \checkmark$$

Q.5. Seasonal variation Short Note $\rightarrow$ See Q.3

Q.6. Semi Supervised Learning Strategy.

$\hookrightarrow$ Machine Learning approach that uses:

- a small amount of labelled data.
- a large amount of unlabelled data.

It combines the advantage of both - Supervised & unsupervised Learning.

Step-1 Collect data

Dataset contains labelled data + unlabelled data.

$\downarrow$
Expensive

$\downarrow$
easily available.

Step-2 Train Initial Model.

The algorithm 1st learns from the small labelled dataset.

Model = $f(x)$ x = input data
 $f(x)$ = predicted o/p.

Step-3 Predict labels for unlabelled data

The trained Model predicts labels for unlabelled data. These predicted labels are called pseudo-labels.

Step-4 Select High Confidence Predictions

Only predictions with high confidence are selected. Low confidences are ignored to reduce errors.

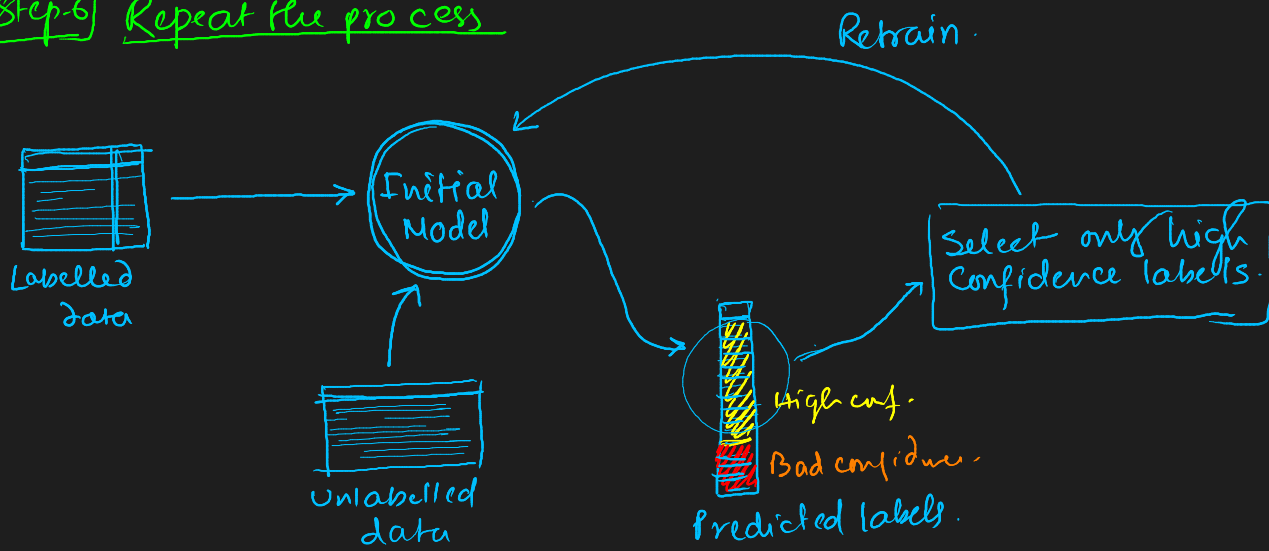
Step-5 Retrain the Model.

The newly labelled data is added to the original labelled dataset. Now the model trains again using -

- original labelled data.
- pseudo labelled data.

This improves learning accuracy.

Step 6 Repeat the process



Q.7. Semi Supervised vs. Reinforcement Learning.

Semi Supervised

- Use both labelled/unlabelled data.
- Combⁿ of supervised + unsupervised.
- Labelled data + unlabelled data
(small) (Large)
- Goal - Improve prediction acc.
- Feedback - use labels + pseudo-labels.
- O/p - Predicted class/value.

Reinforcement Learning

- Agent learns by interacting with env.
- Learning through Reward & Penalty.
- No labelled data Required.
- Maximize cumulative Reward.
- use Reward signal.
- Best action or policy.

Q.8. Advantages & Disadvantages of RL.

Advantages.

- ① Learns through interaction & Experience.
- ② Improves performance continuously.
- ③ Handles complex decision Making problems.
- ④ Works well in dynamic env.
- ⑤ No labelled dataset required.
- ⑥ Can achieve long term optimization.
- ⑦ Useful for Robotics & Game AI.
- ⑧ Adapts automatically to changing conditions.

Disadvantages:

- ① Requires Large training time.
- ② Needs huge computational Resources.
- ③ Explorations may lead to wrong actions.
- ④ Reward design can be difficult.
- ⑤ Training can be unstable.
- ⑥ Not suitable for all problems.
- ⑦ Requires many interactions with environment.
- ⑧ Can be expensive for Real world applications.

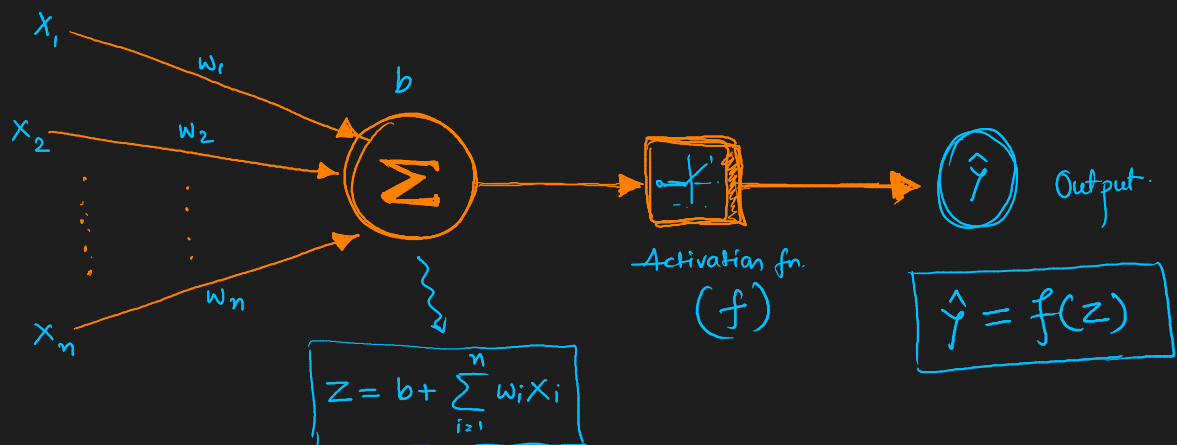
Q.9. Architecture of ANN.

Artificial Neural Network is a computational model inspired by human brain.

The Building Block of an ANN

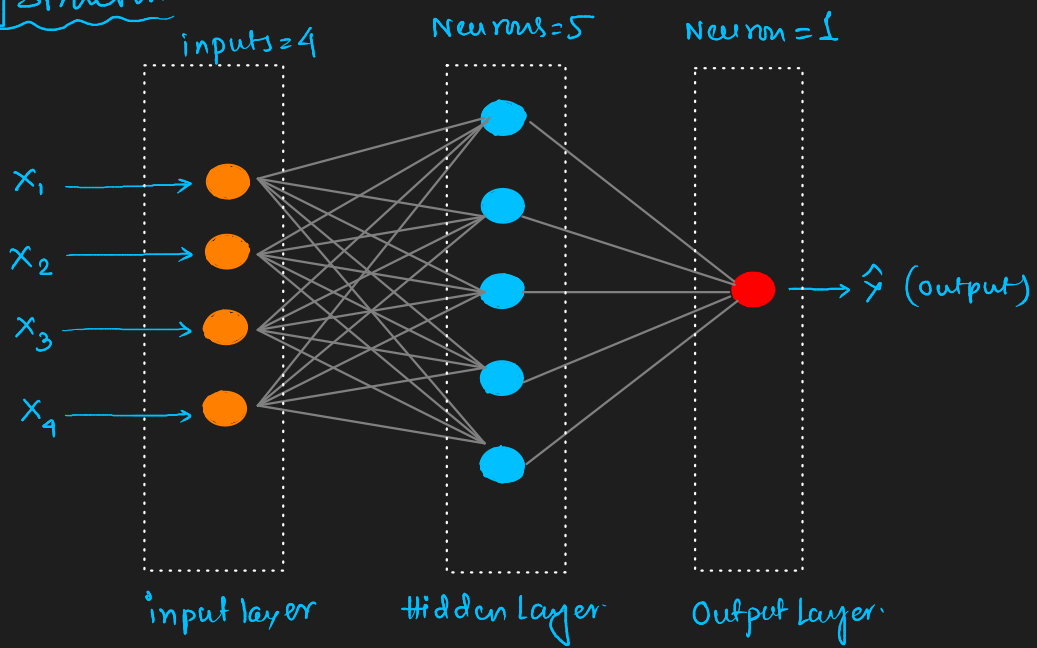
Neuron or Perceptron

A Neuron Structure / Perceptron



ANN $\rightarrow$ Consists of interconnected neurons organized into layers.

A small ANN Structure



the previous ANN architecture is a simple architecture having.

4 inputs $\rightarrow$ 1 hidden layer $\rightarrow$ 1 node output layer.
(5 nodes)

$$\text{Total No. of Params} = (4 \times 5 + 5) + (5 \times 1 + 1) = \boxed{31}$$

inputs

nodes in hidden layer.

5 biases for 5 nodes in hidden layer.

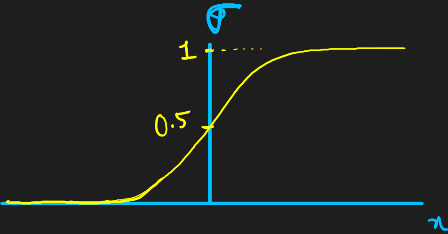
inputs for output layer.

nodes in o/p layer.
o/p layer bias.

Different Types of Activation Fn.

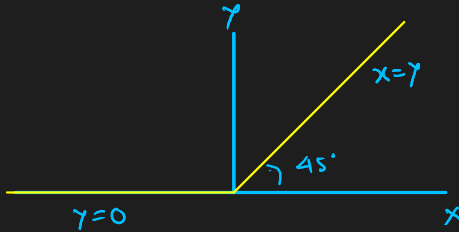
① Sigmoid.

$$\sigma(x) = \frac{1}{1+e^{-x}}$$



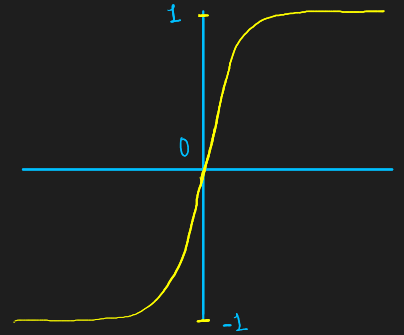
② ReLU

$$f(x) = \max(0, x)$$



③ Tanh

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$



Forward Propagation



$$Z = \sum wx + b$$

$$a = f(z)$$

Q.10 Role of Activation fn in ANN.

- ① Introduces Non Linearity into ANN. (to capture complex patterns)
- ② Decides whether a neuron activates or not.
- ③ Helps learn complex patterns.
- ④ Transforms weighted sum into output.
- ⑤ Enables DL Models to work effectively.
- ⑥ Helps create decision boundaries.
- ⑦ Controls information flow in the Network.
- ⑧ Improves prediction Accuracy.

Q.11 Explain various types of Activation fn. (see Q.9)

Q.12. What are Feed Forward Network? (FFN).

Simplest type of ANN. (IP $\rightarrow$ hidden $\rightarrow$ OP)
(see Q.9).

Q.13. Define Deep Learning. Name few DL Algos. DL vs. ML.

DL $\rightarrow$ Subset of ML that uses Multi Layered ANN to learn complex patterns from large amount of data.

• It automatically extracts features and improves learning through multiple hidden layers.



DL Algos $\rightarrow$

- ANN (Artificial Neural Network)
- CNN (Convolutional Neural Network)
- RNN (Recurrent Neural Network)
- LSTM (Long Short Term Memory)
- Autoencoders
- GANs (Generative Adversarial Networks)

ML

- ① Subset of AI ($ML \subset AI$)
- ② Requires Manual Feature Extraction
- ③ Well for smaller dataset.
- ④ Simple Models.
- ⑤ Faster Training
- ⑥ Req. less comp. power.
- ⑦ Suitable for structured data.
- ⑧ Eg: DT, SVM.

DL

- ① Subset of ML ($DL \subset ML$)
- ② Automatically Extracts features.
- ③ Req. Large dataset.
- ④ Uses deep NN.
- ⑤ Requires a long time
- ⑥ Req. large comp. power.
- ⑦ Best for unstructured data.
- ⑧ Eg: CNN, RNN.